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HARDWARE MINI PROJECT REPORT:

MULTI-RANGE DC VOLTAGE MEASUREMENT CIRCUIT USING OP-AMP

SUBMITTED BY,

KISHORE KUMAR A (2023505018)
DRAVID ANAND D (2023505021)

UNDER THE GUIDANCE OF,

Dr. SABITHA RAMAKRISHNAN
PROFESSOR
DEPARTMENT OF INSTRUMENTATION ENGINEERING

MADRAS INSTITUTE OF TECHNOLOGY CAMPUS

ANNA UNIVERSITY

CHROMPET, CHENNAI -- 600044

MULTI-RANGE DC VOLTAGE MEASUREMENT CIRCUIT USING OP-AMP (IC 741)

1. AIM

To design and implement a multi-range DC voltage measurement circuit using an operational amplifier (IC 741) in non-inverting configuration with negative feedback, providing selectable measurement ranges of 1 V, 5 V, 10 V and 13 V, and to validate the circuit through simulation and hardware testing.

2. EQUIPMENTS AND COMPONENTS REQUIRED

S. No.	Component	Value / Specification	Quantity
1.	Operational Amplifier	IC 741	1
2.	Potentiometer (RV5)	10 kΩ	1
3.	Resistor (R2)	15 kΩ	1
4.	Feedback Resistor (R1)	10 kΩ	1
5.	Gain-setting Resistors (Through Rotary Switch): R6 / R3 / R4 / R5	1 kΩ / 5 kΩ / 10 kΩ / 13 kΩ	1 each
6.	Rotary Selector Switch (SW1)	4-position	1
7.	Analog Voltmeter	0-20 V DC	1
8.	DC Power Supply	±15 V, 1 A	1
9.	Breadboard / PCB	—	1
10.	Connecting Wires	—	As required

3. CIRCUIT DIAGRAM

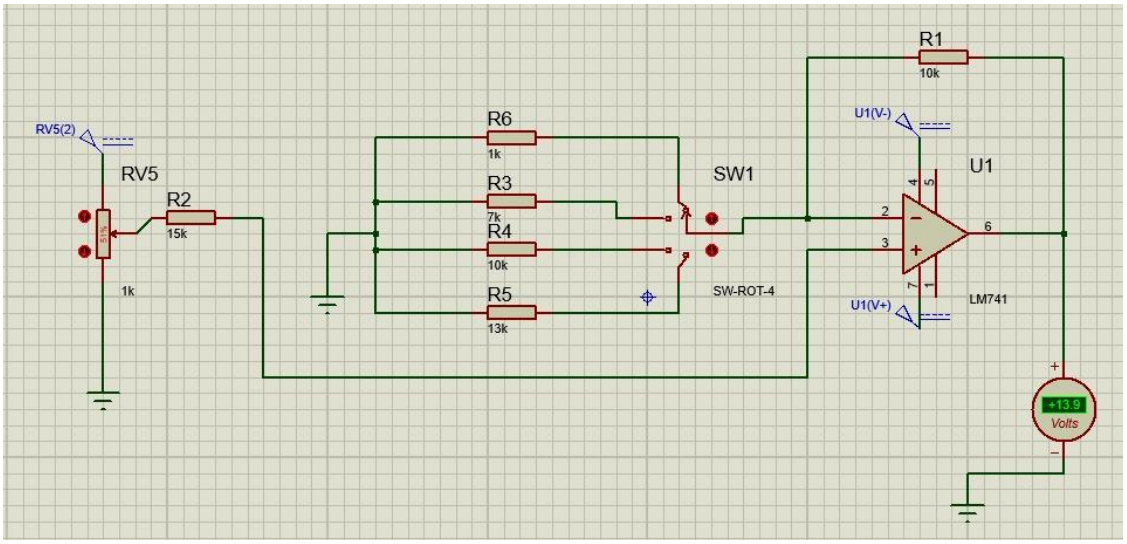

Figure 1: Multi-range DC voltage measurement circuit using LM741.

4. THEORY

The circuit uses an operational amplifier in non-inverting configuration with negative feedback. The input DC voltage is applied to the non-inverting input through R2. The inverting input is connected to the output through feedback resistor R1, and to ground through a selectable resistor (R6 / R3 / R4 / R5) using a rotary switch.

The closed-loop voltage gain of a non-inverting amplifier is,

$$Av = 1 + Rf / Rg$$

where, Rf = Feedback resistor (R1)

Rg = Selected resistor (R6, R3, R4 or R5)

The output voltage is given by,

$$Vout = Av \times Vin$$

5. ROLE OF COMPONENTS

- IC 741: Provides voltage amplification with high input impedance and negative feedback for stable operation.
- R1 (10 kΩ): Feedback resistor providing negative feedback from output to inverting input.
- R6, R3, R4, R5: Gain-setting resistors selected by rotary switch to obtain different voltage gains.
- SW1 (Rotary Switch): Allows selection of one resistor at a time to set the required measurement range.
- RV5 (1 kΩ) Potentiometer: Provides adjustable DC input voltage for testing and calibration.

- R2 (15 kΩ): Limits current from the source and provides proper input resistance to the non-inverting terminal.
- Analog Voltmeter: Measures the amplified output voltage.
- ±15 V Supply: Powers the LM741 for proper operation.

6. CIRCUIT LOGIC AND OPERATION

1. The DC input voltage is applied using potentiometer RV5.
2. The input is fed to the non-inverting terminal (pin 3) of IC 741 through R2.
3. The output is fed back to the inverting terminal (pin 2) through R1 (10 kΩ).
4. The inverting terminal is connected to ground through one of the resistors (1 kΩ, 7 kΩ, 10 kΩ or 13 kΩ) via rotary switch SW1.
5. Due to negative feedback, the op-amp provides a closed-loop gain of Av = 1 + Rf / Rg.
6. Selecting different resistors changes the gain and hence the output voltage.
7. The output voltage is measured using the analog voltmeter.
8. Input voltage is varied and readings are recorded for each selected range.

7. THEORETICAL CALCULATION

Given: Rf = R1 = 10 kΩ

Gain for each selected resistor:

Range (No.)	Selected Resistor (Rg)	Gain (Av) = 1 + Rf/Rg	Relation Vout = Av × Vin
1 (1 V range)	Rg = R6 = 1 kΩ	1 + 10/1 = 11.00	Vout = 11.00 Vin
2 (5 V range)	Rg = R3 = 7 kΩ	1 + 10/7 = 2.43	Vout = 2.43 Vin
3 (10 V range)	Rg = R4 = 10 kΩ	1 + 10/10 = 2.00	Vout = 2.00 Vin
4 (13 V range)	Rg = R5 = 13 kΩ	1 + 10/13 = 1.77	Vout = 1.77 Vin

Note: The above gains determine the output level for a given input voltage. The meter scale is calibrated for each range to read the actual input voltage (1 V, 5 V, 10 V and 13 V).

8. CALIBRATION PROCEDURE

1. Connect the circuit as per the diagram and give ±15 V supply to IC 741.
2. Set the rotary switch to a required range.
3. Set the input voltage to zero using RV5.
4. Apply a known DC input voltage.
5. Observe the output reading on the voltmeter.
6. Adjust RV5 and note the output for different input values.
7. Repeat the procedure for all four ranges.

8. Calibrate the meter scale so that full-scale deflection corresponds to 1 V, 5 V, 10 V or 13 V as per the selected range.

9. SIMULATION RESULTS (Proteus)

The circuit was simulated in Proteus. The output varied linearly with input voltage for all selected resistors. The gains obtained matched the theoretical values.

Range	Selected Resistor (Rg)	Input Voltage (Vin)	Simulated Output (Vout)	Calculated Gain (Vout/Vin)
1 V range	1 kΩ	0.100 V	1.10 V	11.00
5 V range	7 kΩ	1.000 V	2.43 V	2.43
10 V range	10 kΩ	1.000 V	2.00 V	2.00
13 V range	13 kΩ	1.000 V	1.77 V	1.77

10. HARDWARE RESULTS

The circuit was built on breadboard and tested. The measured readings were in good agreement with the theoretical values.

Range	Applied Input Voltage (Vin)	Measured Output (Vout)	Reference Output (Calculated)	% Error
1 V range	0.000 V	1.09 V	1.10 V	0.91 %
5 V range	1.000 V	2.41 V	2.43 V	0.82 %
10 V range	1.000 V	1.98 V	2.00 V	1.00 %
13 V range	1.000 V	1.77 V	1.77 V	1.13 %

11. ADVANTAGES

- Simple and low-cost design.
- Selectable measurement ranges using rotary switch.
- High input impedance and good stability due to negative feedback.
- Easy calibration and wide range of applications.

12. LIMITATIONS

- Accuracy depends on resistor tolerance and op-amp offset.
- Output limited by supply voltage and op-amp swing.
- Manual range selection required.
- Suitable only for DC voltage measurement.

13. APPLICATIONS

- Low-voltage DC measurement systems.
- Educational and laboratory measurement setups.
- Instrumentation and data acquisition systems.
- Analog signal conditioning applications.

14. INFERENCE

The designed multi-range DC voltage measurement circuit using LM741 op-amp operates satisfactorily. Different feedback resistor selections produce different gains, enabling measurement of DC voltages in the ranges of 1 V, 5 V, 10 V and 13 V. Simulation and hardware results confirm the validity of the design.

15. RESULT

Thus, the multi-range DC voltage measurement circuit using LM741 op-amp was successfully designed, implemented and tested. The circuit provides accurate voltage measurement in the ranges of 1 V, 5 V, 10 V and 13 V.